

Rim Analysis Tool

User Guide

Purpose: The program lets you manually place an elliptical or circular region of interest (ROI) on a vesicle membrane, sample fluorescence intensity around the selected rim, inspect the angular intensity profile, and save the measured data and plot. Viewer zoom, contrast and brightness adjustments are display-only and do not alter the underlying measurement values.

Quick start

Step	Action
1	Start the program and click Load Image.
2	Use the mouse wheel to zoom at the cursor and middle-mouse drag to pan.
3	Draw an ellipse with the left mouse button. Hold Shift while drawing to create a circle.
4	Refine the ROI using the center and edge handles.
5	Choose rim thickness and any optional display/plot settings.
6	Click Process Image to calculate and display the intensity profile.
7	Use Save CSV and/or Save Plot Image to export the result.

1. Installation on a new Windows workstation

A Python virtual environment is recommended so the program and its dependencies remain isolated from other Python software on the workstation.

1.1 Install Python

Install a normal standalone Python 3.11 or 3.12 distribution.

```
py --version
```

If more than one Python installation exists, list them with:

```
py -0p
```

1.2 Create and activate a virtual environment

Open PowerShell in the folder that contains the script, then run:

```
py -3.12 -m venv .venv
.\.venv\Scripts\Activate.ps1
```

PowerShell execution policy: If activation is blocked, either use Command Prompt or allow locally created scripts for the current user with: Set-ExecutionPolicy -Scope CurrentUser RemoteSigned

1.3 Install the required packages

```
python -m pip install --upgrade pip
python -m pip install numpy matplotlib opencv-python pillow
```

Tkinter is included with the standard Windows Python installer. You can test it with:

```
python -m tkinter
```

1.4 Start the program

```
python RimAnalysisTool.py
```

2. Program interface

The scrollable control panel is on the left; the image viewer occupies the area on the right.

Image. Load Image opens a PNG, JPEG, TIFF or TIF file. The loaded filename is shown below the button.

Rim Selection. Contains the selection color, rim thickness, curve smoothing option and plot color.

Viewer Display. Contains Brightness and Contrast sliders plus Auto Contrast, Reset B/C and Fit Image. These settings affect only what you see in the viewer.

Analysis. Process Image calculates the profile. Save CSV and Save Plot Image remain disabled until a valid analysis has been processed.

Controls & Shortcuts. A reference describing mouse and keyboard controls.

Important: Brightness, contrast, automatic contrast, and zoom do not modify the source image used for quantitative analysis. Intensity values are sampled from the untouched image loaded from disk.

3. Loading and viewing an image

3.1 Load an image

1. Click Load Image.
2. Select a supported image file.
3. The image is displayed and automatically fitted to the viewer.
4. Automatic display contrast is applied by default to make dim fluorescent vesicles easier to see.

3.2 Viewer display controls

Control	Function
Brightness	Raises or lowers the displayed intensity after the base display mapping.
Contrast	Changes displayed contrast. The wide range is useful for low-intensity vesicles.
Auto Contrast	Reapplies automatic fluorescence-friendly contrast enhancement.
Reset B/C	Returns to the original, unenhanced display and resets brightness/contrast to neutral values.
Fit Image	Fits the full image into the available viewer area.

3.3 Zoom and pan

- Mouse wheel: zooms in or out around the current cursor position.
- Right-mouse drag: pans a zoomed image.
- + / -: zoom in or out around the center of the visible viewer.
- 0: return to 1:1 zoom.
- F: fit the image to the viewer.

4. Selecting the vesicle rim

4.1 Draw an ellipse or circle

- Left-click and drag across the vesicle to create an elliptical ROI.
- Hold Shift while dragging to constrain the initial ROI to a circle.

4.2 Adjust the ROI with handles

Five handles are available: one at the center and one at each of the left, right, top and bottom edges.

Action	Effect
Drag center handle	Move the complete ROI without changing its size.
Drag an edge handle	Adjust only that side of the ellipse.
Shift + edge handle	Adjust opposite sides symmetrically along that axis.
Shift + Alt + edge handle	Resize uniformly in all directions as a circle.
D	Clear the current selection so you can redraw it.

5. Analysis settings

5.1 Selection color

Choose white or grey for the ROI overlay. This is a viewer preference only.

5.2 Rim Thickness

Rim Thickness controls the width of the membrane region sampled around the selected ellipse. The displayed ROI line changes thickness as you move the slider, giving immediate visual feedback. The setting is recorded in the analysis log.

5.3 Smooth Curve

When enabled, a simple 5-point moving average is applied to the measured angular intensity profile and to the exported intensity values. In other words, each point in the plotted profile is averaged with nearby points to reduce small point-to-point fluctuations and make broader intensity trends around the vesicle membrane easier to see.

5.4 Plot Color

Select the line color used for the intensity plot. Available options include blue, red, green, black, orange and purple.

6. Processing and saving a measurement

6.1 Process Image

After placing the ROI, click Process Image. The program samples fluorescence intensity around 360 angular positions along the selected rim and opens the intensity-versus-angle plot.

Important: Save CSV and Save Plot Image are disabled until Process Image has generated a valid result. If you redraw or resize the ROI, or change Rim Thickness, the previous result is invalidated and the save buttons are disabled again until you reprocess the image.

6.2 Save CSV

Save CSV writes a two-column file containing Angle (degrees) and Intensity. A text analysis log is also written using the same base filename followed by _log.txt.

The log records the image filename, analysis settings, display settings at the time of saving, viewer zoom and summary statistics such as mean, maximum, minimum and standard deviation.

6.3 Save Plot Image

Save Plot Image exports the current profile as PNG or SVG. PNG output is saved at high resolution.

7. Keyboard and mouse shortcuts

Shortcut / action	Function
Mouse wheel	Zoom at cursor
Middle-drag	Pan image
+ / -	Zoom in / out
0	Reset zoom to 1:1
F	Fit image to viewer
LMB drag	Draw ellipse
Shift + LMB drag	Draw circle
Drag handle	Adjust one side / move center
Shift + handle	Adjust symmetrically
Shift + Alt + handle	Uniform circular adjustment
D	Redraw selection
R	Original display / reset B/C

8. Recommended measurement workflow

1. Load one vesicle image.
2. Use Auto Contrast if needed to make the membrane easy to identify.
3. Zoom to the vesicle and pan as needed.
4. Draw the ellipse/circle and refine it with the handles so the ROI follows the membrane rim.
5. Set Rim Thickness to the desired sampling width.
6. Click Process Image and inspect the resulting intensity profile.
7. Save the CSV and/or plot image.
8. For the next vesicle, load a new image and repeat.